25900747 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCBTS100 - NEW BIOLOGY FOR THE NEW CENTURY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Haematoxylin and Eosin are [K] / [CO3]

   a). Enzymes b). Antibodies c). Dyes d). None of the above

2. What is the significance of the cell cycle checkpoints? [U] / [CO1]

   a). To ensure DNA replication occurs b). To regulate the progression of the cell cycle c). To promote cell division d). To initiate apoptosis

3. Which technique resembles taking a molecular photograph? [K] / [CO3]

   a). X ray crystallography b). NMR c). Radioisotope technique d). none of the above

4. Using flow cytometry we can know [U] / [CO2, CO3, CO7]

   a). Cell size b). Cell shape c). Cell type d). All of the above

5. The ability of an ecosystem to recover from disturbances is known as: [K] / [CO4]

a). Resilience b). Resistance c). Adaptation d). Stability

6. Which of the following is a function of the cell membrane? [U] / [CO1]

a). Energy production b). Genetic material storage c). Regulation of material transport d). Protein synthesis

7. Which of the following best describes a prokaryotic cell? [U] / [CO2]

a). Contains a nucleus b). Lacks membrane-bound organelles c). Is always multicellular d). Has a complex structure

8. Which of the following is a characteristic of organisms in the same population? [U] / [CO2]

a). They share the same habitat. b). They belong to different species c). They reproduce together. d). They have different genetic makeups.

9. If a DNA sequence changes from A-T-C to A-G-C, what type of mutation has occurred? [A] / [CO4]

a). Insertion b). Deletion c). Substitution d). Duplication

10. Genome is [An] / [CO3, CO7]

a). a random collection of genes b). The set of genes encoded at the time of birth c). The entire DNA content present in one cell d). The entire DNA content present in all cells

[10x1 = 10]

Part B

Short Answer Type Questions

Answer any **10** Questions

Each Question carries **2** marks

11. Write down the principle and application of mass spectrometry [U] / [CO3]

12. How did the study of antibiotic resistance illustrate the principles of evolution? [U] / [CO6]

13. Discuss on visualisation and live imaging in disease diagnosis and therapeutics. [A] / [CO5, CO7]

14. What is speciation, and what are the mechanisms that can lead to it? [U] / [CO6]

15. Briefly explain the concept of independent assortment. [U] / [CO4]

16. What is one limitation of Whittaker's five-kingdom classification system? [An] / [CO6]

17. How is the production of monoclonal antibodies carried out? [A] / [CO7]

18. What is the function of the nucleus in eukaryotic cells? [U] / [CO1]

19. Define the term 'genetic code.' [K] / [CO1]

20. What is the role of spindle fibers during cell division? [U] / [CO2]

21. What is the significance of biodiversity hotspots? [U] / [CO3]

22. Expand NMR and mention its principle. [K] / [CO3]

[10x2 = 20]

Part C

Short Essay Type Questions

Answer any **5** Questions

Each Question carries **4** marks

23. Write in detail on the steps in recombinant DNA technology [K] / [CO4]

24. Identify the four main types of biomolecules. Briefly describe the primary function of each type in living organisms. [K] / [CO3]

25. Discuss the principles of inheritance as proposed by Gregor Mendel and how they apply to modern genetics. [U] / [CO4]

26. Give an account on the challenges and applications of biotechnology and nanotechnology in the modern world [A] / [CO6]

27. Explain the concept of the central dogma of molecular biology. [U] / [CO1]

28. Explain the concept of independent assortment and how it contributes to genetic diversity. [U] / [CO4]

29. Write an essay on biological databases. [U] / [CO5]

[5x4 = 20]